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ODD END SEMESTER EXAMINATION – 2025

Name of the Course: B.Tech (ECE)

Semester: III

Name of the Paper: Electronic Devices and Circuits

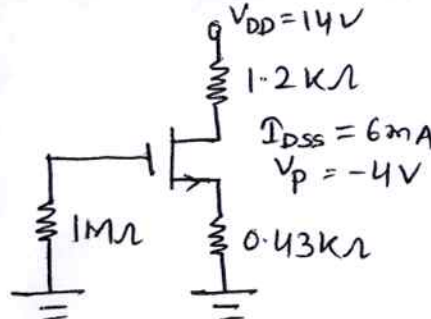
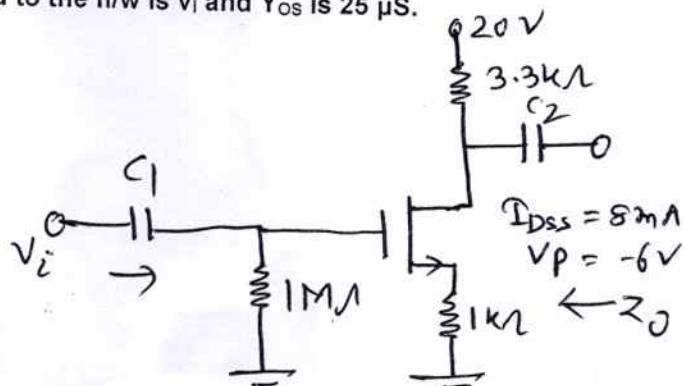
Paper Code: **TEC 301**

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Explain different types of semiconductors. Discuss the concept of electron and hole flow in it.	CO1, CO3
(b)	Discuss the working of the NPN bipolar junction transistor with a proper diagram.	
(c)	Explain the voltage divider bias circuit for BJT Biasing and write the condition for the operation of the transistor in the active region.	
Q2	(20 marks)	
(a)	What is RC-coupled transistor amplifier? Discuss it with a proper diagram and also plot the frequency response curve.	CO2, CO3
(b)	Discuss the transfer and output characteristics of n-channel enhancement MOSFET.	
(c)	For the self-bias configuration, determine the following parameters (a) I_{DQ} and V_{GSQ} (b) V_{DS} and V_D	
		
Q3	(20 marks)	
(a)	Explain MOSFET as an amplifier, discuss its small signal model, and find the expression for transconductance (g_m).	CO3, CO5
(b)	Draw the a.c equivalent circuit of the self-bias configuration of N-type depletion MOSFET and find the value of voltage gain.	
(c)	The self-bias configuration of the given figure has an operating point defined by $V_{GSQ} = -2.6\text{ V}$ & $I_{DQ} = 2.6\text{ mA}$, with $I_{DSS} = 8\text{ mA}$ and $V_P = -6\text{ V}$. The input signal applied to the n/w is v_i and Y_{OS} is $25\text{ }\mu\text{S}$. (a) Determine g_m . (b) find r_d (c) find Z_i (d) calculate Z_o (e) calculate A_v .	
		

Q4	(20 marks)	CO3, CO4
(a)	Describe the different properties of negative feedback & also show that due to negative feedback, stability and the input resistance of the voltage amplifier increase.	
(b)	How oscillator is different from an amplifier? Discuss different types of oscillators.	
(c)	Draw the circuit of the Hartley oscillator and explain its working. Derive an expression for frequency of oscillation.	
Q5	(20 marks)	CO5, CO6
(a)	Describe the working of the Class A power amplifier in detail with a relevant diagram.	
(b)	Describe the working of the Class B push pull power amplifier.	
(c)	A class A, CE amplifier operates from $V_{cc} = 20\text{ V}$, flows a no signal current of 5 A, and feeds a load of $40\ \Omega$ through a step-up transformer of $n_2/n_1 = 3.16$. Find (i) whether the amplifier is properly matched for maximum power transfer. (ii) maximum a.c signal power output. (iii) maximum d.c power input. (iv) conversion efficiency at maximum signal O/P.	